Bounded-memory Boolean Macaulay passes

Execution record, 18 October 2026. Items 7 and 9; public synthetic inputs.

Eight degree-4 passes on actual chained S3 Weil-descended systems completed. Four small controls agreed exactly with an independent dense bit-matrix RREF. The larger passes remain independently unverified. This is a fixed-degree row-reduction result, not full F4, iterative closure, solving degree, or an ECDLP cost result.

n	ℓ	Variables	Rows / rank	Dense RREF control
5	2	18	1,050 / 987	Exact equality, both policies
9	3	30	4,752 / 4,648	Exact equality, both policies
15	5	50	20,670 / 20,490	256 MiB reference limit refused
17	6	58	31,110 / 30,888	Open on diagnostic continuation



One target [17]G per shape and policy; K_1 curves over $\mathbf{F}_{2^n}$, polynomial-basis subspaces, $m = 4$, $D = 4$. Graph values are native recorded counters. Large-case ranks are engine observations, not independent certificates. Sparse-entry counters cover committed checkpoint history and exclude rolled-back suffix work; they establish no full-cost speed comparison.

The n=5 zero-cache and row-31 checkpoint/restart controls returned the same independently certified RREF. One hundred seeded native arithmetic/oracle controls passed. Truncated streams, inadequate output disk, restart input SHA mismatch, and the n=19/ell=7 layout requiring 66 variables failed closed. The current exporter is limited to 64-bit monomials.

The original native-v1 launches exited 127 because /usr/bin/time was unavailable; no scientific binaries ran in that attempt. A native wait4 launcher replaced it and enforced a 536,870,912-byte child address-space ceiling. Initial n=15 runs reached EOF before the dense oracle's 256 MiB allocation guard refused them. Initial n=17 runs reached their 64 MiB trace cap. Copies of checkpoints/journals were continued in fresh directories, preserving original failures.

Scope, accounting and evidence

The pivot cache has an explicit 1,048,576-byte row-payload limit with LRU eviction. Index/LRU metadata and temporary rows add overhead; the process-wide address-space limit bounds total address space. Journal and trace caps are 1 GiB and 64 MiB per phase. Row-size guards allow 16 MiB of terms; temporary XOR vector reservations may use 32 MiB before the guard. The reducer is serial and disk-backed. There is no GPU, distributed executor, or incumbent-oracle integration.

Per-process CPU, wall time and peak RSS are recorded separately for every launch in resources.json files. The host was shared (isolation tier 0). Initial n=15 first/minimum-weight runs recorded 26,932/27,048 KiB peak RSS. Build, export, generation, failed attempts, verification, copied state and resumed reductions are separate phases. Journal counters omit index reconstruction/input reads and rolled-back suffix work; process resource records retain their executed costs. No matched external engine or full pipeline/rho comparison was run.

The policy hooks allow only choosing between rows with the same leading monomial. First-pivot and minimum-weight controls log lead, row index, incoming and pivot weight, replacement action, and feature/decision timer counts. No learned model, training/calibration/heldout split, charged training experiment, learned column policy or heldout transfer result exists in this deliverable.

Checkpoints flush journal data before atomically renaming metadata, binding input SHA256, policy, cache size, next input offset, journal boundary and counters. The tests establish the stated process restart and input-identity checks; power-loss/fsync durability and arbitrary journal-corruption handling remain open.

Item 7 remains partial: canonical frontier ell=6/7 and guess-count ell=8/9 closure trials, full F4/MutantXL scheduling, certified large-case replay and distributed/GPU kernels remain outstanding. The selected n=17/ell=6 cell is a diagnostic, not a substitution for a canonical frozen frontier requirement. Item 9 has deterministic legal hooks only. No solving-degree, deployed-security or fastest-engine claim is made. Canonical leaderboard and solving-degree plots stay unchanged.

Reproducible evidence. The study directory is `research/ideas_execution_20261018/bounded_memory/`. Frozen protocols precede their respective trial scripts. Runs native-v1, native-v2 and large-continuation-v1 retain exact argv, UTC times, exits, raw stderr, resources, input/source/binary/dependency digests, checkpoints and traces. Input metadata binds field structure, target coordinates, subgroup order and subspace basis. Gzip archives preserve raw record contents; SHA256 manifests refer to their uncompressed basenames. Full observations and links are in REPORT.md and README.md.

Sources. Pinned source base: `e063d98475589d9aba7ae265a8cba830927e3156`, with root-agent library build repairs. The ANF builder and row family are from `src/cryptanalysis/koblitz_groebner.rs`; the existing sparse control context is `sparse_macaulay.rs`. Full F4 semantics in `pq_f4_f2.rs` are explicitly outside this standalone reducer. No library core is edited here.